The Geometry of Consolidation
Supplementary Material

Anirudh Bharadwaj Vangara[1,2] and Ashwin Gopinath*[1,3]

[1]Sentra, 235 2nd Street, San Francisco, CA 94105, USA
[2]Department of Computer Engineering, University of Waterloo, Waterloo, ON N2L 3G1, Canada
[3]Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

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Contents

S1 Scope of this supplementary

This document is a companion to the main paper; it does not restate its narrative. It collects (i) per-cell tables for every experimental grid referenced in the main paper, (ii) an expanded E8 encoder grid with two additional figures, (iii) the full c_1 calibration table at every (d_{eff}, θ) row of

*ashwin@sentra.app

the E1 grid, (iv) a run manifest mapping every main-paper figure to its source parquet, and (v) the exact reproduction recipe. The proof of the Duality Theorem and the statement of Definition 1 (d_{eff} as the participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$) live in the main paper's Methods section and are not duplicated here.

What is *not* in this supplementary. No new theorems. No additional experiments beyond those referenced in the main paper. No claims about c_1 or the regime boundary that are not already stated quantitatively in §3 and §4 of the main paper.

S2 Expanded E8 encoder-strategy grid

The main paper reports E8 on a 5-strategy slice (6 encoders $\times$ 5 strategies $\times$ 3 seeds = 90 cells). For completeness we re-ran E8 on the full $6 \times 11 \times 3 = 198$-cell grid that also sweeps GAC's threshold $\theta \in \{0.6, 0.7, 0.8, 0.9\}$, four SELECTIVE_PRUNE keep-ratios $(0.50, 0.25, 0.10)$, and the *no_consolidation* identity ceiling. Two figures summarise the expanded data.

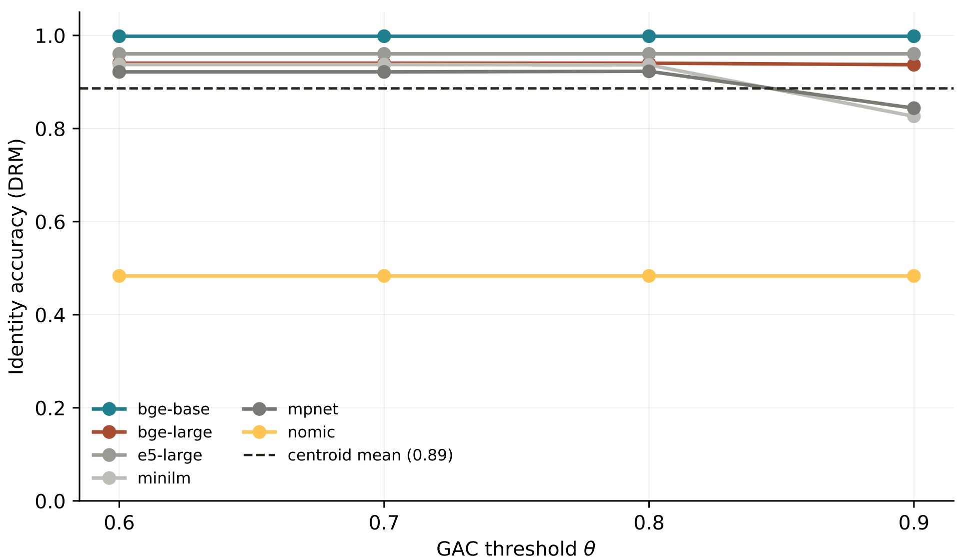

Figure S1: E8 expanded: GAC threshold θ sweep on DRM. Each line is one encoder; the dashed line is the mean centroid baseline. Identity accuracy is approximately flat across $\theta \in [0.6, 0.9]$ for every encoder — consistent with the main paper's finding that tight DRM clusters fall in the regime where the centroid branch of the GAC router dominates, making the router insensitive to θ.

S3 E1 strategy sweep and identity–coverage scatter

Two auxiliary plots that did not fit in the main text.

S4 Complete c_1 calibration table

Table S1 gives the per-cell c_1 calibration row by row across the E1 grid. The aggregate numbers reported in §4 of the main paper ($c_1^{95} \approx 0.046$ on 12,400 tight cells; $c_1^{95} \approx 4.61 \times 10^{10}$ on 3,600 spread cells; global $c_1^{95} \approx 4.60 \times 10^{6}$ on all 16,000 cells) are the $p95$ quantiles of the per-row c_1 distribution

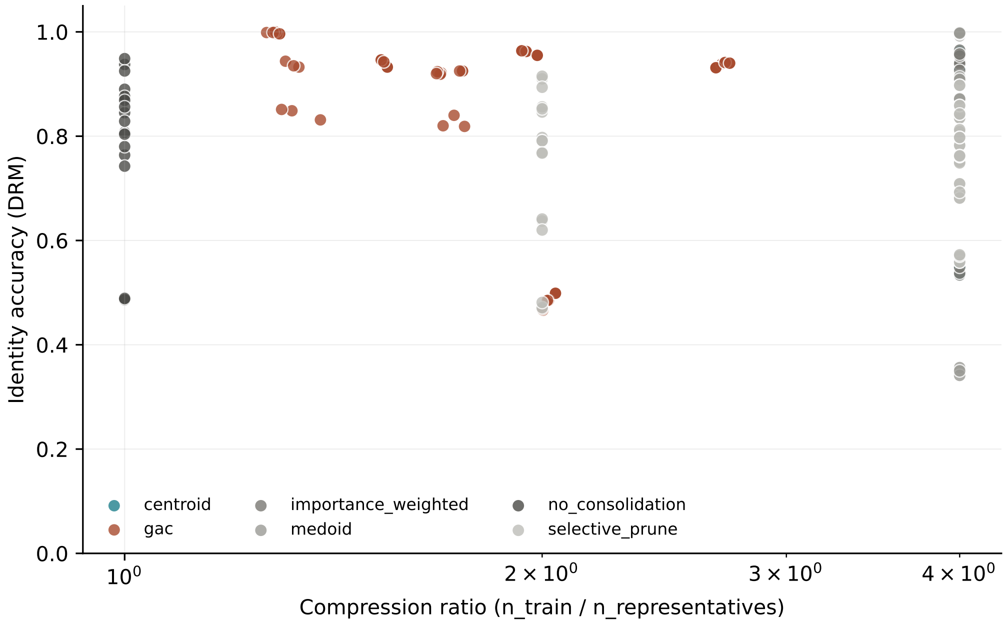

Figure S2: E8 expanded: compression–identity frontier across the full $6 \times 11 \times 3$ grid. Each point is one seed. The horizontal axis is the empirical compression ratio $n_{\text{train}}/n_{\text{representatives}}$; the right edge is *no_consolidation* by construction. Centroid and medoid occupy the upper-right (compressed + accurate) corner on every encoder; SELECTIVE_PRUNE degrades continuously as the keep-ratio shrinks.

restricted to the relevant regime. The full per-$(d_{\text{eff,target}}, \theta)$ breakdown below lets a reader audit the regime boundary and verify that the c_1 explosion is localised to the extreme-coherence corner (high θ, low d_{eff}), not an artifact of outlier rows.

S5 Per-cell experiment tables

These tables enumerate every row of the main experimental grids so that aggregated statistics in the main paper can be reproduced by inspection.

Table S2: E2 per-cell results (strategy sweep across 6 corpora, 6 encoders, 4 clusterers). (showing first 60 rows; full CSV in repo)

corpus	model	clusterer	strategy	seed	n_train	compression	identity_accuracy
arxiv_titles	bge-large	kmeans	centroid	0	25545	2322.27	0.754
arxiv_titles	bge-large	hdbscan	centroid	0	25545	2322.27	0.754
arxiv_titles	bge-large	hdbscan	gac	0	25545	2.00	0.582
arxiv_titles	bge-large	kmeans	gac	0	25545	2.00	0.582
arxiv_titles	bge-large	hdbscan	gac	0	25545	2.00	0.582
arxiv_titles	bge-large	kmeans	gac	0	25545	2.00	0.582
arxiv_titles	bge-large	kmeans	importance_weighted	0	25545	2322.27	0.754
arxiv_titles	bge-large	hdbscan	importance_weighted	0	25545	2322.27	0.754

(continued)

corpus	model	clusterer	strategy	seed	n_train	compression	identity_accuracy
arxiv_titles	bge-large	kmeans	medoid	0	25545	2322.27	0.611
arxiv_titles	bge-large	hdbscan	medoid	0	25545	2322.27	0.611
arxiv_titles	bge-large	kmeans	selective_prune	0	25545	2.00	0.582
arxiv_titles	bge-large	hdbscan	selective_prune	0	25545	2.00	0.582
arxiv_titles	bge-large	kmeans	selective_prune	0	25545	4.00	0.442
arxiv_titles	bge-large	hdbscan	selective_prune	0	25545	4.00	0.442
c4_random	bge-large	kmeans	centroid	0	8787	17.57	0.713
c4_random	bge-large	kmeans	centroid	0	8787	17.57	0.012
c4_random	bge-large	kmeans	centroid	0	8787	17.57	0.002
c4_random	bge-large	kmeans	centroid	0	8787	17.57	0.002
c4_random	bge-large	kmeans	gac	0	8787	4.27	0.626
c4_random	bge-large	kmeans	gac	0	8787	4.27	0.006
c4_random	bge-large	kmeans	gac	0	8787	4.27	0.002
c4_random	bge-large	kmeans	gac	0	8787	4.27	0.002
c4_random	bge-large	kmeans	gac	0	8787	2.04	0.312
c4_random	bge-large	kmeans	gac	0	8787	2.04	0.005
c4_random	bge-large	kmeans	gac	0	8787	2.04	0.005
c4_random	bge-large	kmeans	gac	0	8787	2.04	0.004
c4_random	bge-large	kmeans	medoid	0	8787	17.57	0.650
c4_random	bge-large	kmeans	medoid	0	8787	17.57	0.010
c4_random	bge-large	kmeans	medoid	0	8787	17.57	0.003
c4_random	bge-large	kmeans	medoid	0	8787	17.57	0.002
c4_random	bge-large	kmeans	selective_prune	0	8787	4.49	0.206
c4_random	bge-large	kmeans	selective_prune	0	8787	4.49	0.007
c4_random	bge-large	kmeans	selective_prune	0	8787	4.49	0.005
c4_random	bge-large	kmeans	selective_prune	0	8787	4.49	0.004
c4_random	bge-large	kmeans	selective_prune	0	8787	8.08	0.167
c4_random	bge-large	kmeans	selective_prune	0	8787	8.08	0.008
c4_random	bge-large	kmeans	selective_prune	0	8787	8.08	0.007
c4_random	bge-large	kmeans	selective_prune	0	8787	8.08	0.006
drm_templated	bge-large	hdbscan	centroid	0	3200	4.00	0.943
drm_templated	bge-large	kmeans	centroid	0	3200	4.00	0.943
drm_templated	bge-large	kmeans	gac	0	3200	1.55	0.932
drm_templated	bge-large	hdbscan	gac	0	3200	1.55	0.932
drm_templated	bge-large	kmeans	gac	0	3200	1.34	0.932
drm_templated	bge-large	hdbscan	gac	0	3200	1.34	0.932
drm_templated	bge-large	kmeans	importance_weighted	0	3200	4.00	0.941
drm_templated	bge-large	hdbscan	importance_weighted	0	3200	4.00	0.941
drm_templated	bge-large	kmeans	medoid	0	3200	4.00	0.910
drm_templated	bge-large	hdbscan	medoid	0	3200	4.00	0.910
drm_templated	bge-large	kmeans	selective_prune	0	3200	2.00	0.769
drm_templated	bge-large	hdbscan	selective_prune	0	3200	2.00	0.769
drm_templated	bge-large	kmeans	selective_prune	0	3200	4.00	0.761
drm_templated	bge-large	hdbscan	selective_prune	0	3200	4.00	0.761
drm_templated	e5-large	kmeans	centroid	0	3200	4.00	0.965

(continued)

corpus	model	clusterer	strategy	seed	n_train	compression	identity_accuracy
drm_templated	e5-large	hdbscan	centroid	0	3200	4.00	0.965
drm_templated	e5-large	kmeans	gac	0	3200	1.95	0.963
drm_templated	e5-large	hdbscan	gac	0	3200	1.95	0.963
drm_templated	e5-large	kmeans	gac	0	3200	1.95	0.963
drm_templated	e5-large	hdbscan	gac	0	3200	1.95	0.963
drm_templated	e5-large	hdbscan	importance_weighted	0	3200	4.00	0.965
drm_templated	e5-large	kmeans	importance_weighted	0	3200	4.00	0.965

Table S3: E3 per-cell results (learned baselines: PQ, OPQ, LSH, PCA+int8, HNSW-prune). (showing first 60 rows; full CSV in repo)

corpus	family	bytes_per_vec	identity_accuracy	recall@1	recall@10	recall@100
arxiv_titles	hnsw_prune	4096	0.466	0.466	0.872	0.994
arxiv_titles	hnsw_prune	4096	0.518	0.517	0.877	0.995
arxiv_titles	hnsw_prune	4096	0.594	0.594	0.912	0.994
arxiv_titles	hnsw_prune	4096	0.668	0.670	0.941	0.996
arxiv_titles	lsh	1	0.127	0.129	0.693	0.997
arxiv_titles	lsh	4	0.310	0.311	0.834	0.998
arxiv_titles	lsh	16	0.529	0.526	0.935	0.999
arxiv_titles	lsh	64	0.684	0.686	0.953	0.999
arxiv_titles	opq	4	0.679	0.679	0.938	0.995
arxiv_titles	opq	8	0.699	0.699	0.945	0.996
arxiv_titles	opq	16	0.709	0.710	0.956	0.996
arxiv_titles	opq	32	0.710	0.710	0.955	0.996
arxiv_titles	opq	64	0.716	0.720	0.954	0.997
arxiv_titles	pca_int8	2	0.315	0.306	0.867	0.991
arxiv_titles	pca_int8	4	0.450	0.448	0.911	0.993
arxiv_titles	pca_int8	8	0.611	0.615	0.937	0.993
arxiv_titles	pca_int8	16	0.682	0.683	0.946	0.995
arxiv_titles	pca_int8	32	0.696	0.702	0.953	0.995
arxiv_titles	pca_int8	64	0.718	0.717	0.951	0.995
arxiv_titles	pq	4	0.664	0.663	0.911	0.991
arxiv_titles	pq	8	0.692	0.687	0.917	0.990
arxiv_titles	pq	16	0.710	0.709	0.930	0.991
arxiv_titles	pq	32	0.716	0.720	0.937	0.991
arxiv_titles	pq	64	0.731	0.731	0.950	0.994
arxiv_titles	pq	128	0.720	0.723	0.956	0.997
drm_templated	hnsw_prune	4096	0.925	0.925	1.000	1.000
drm_templated	hnsw_prune	4096	0.930	0.930	1.000	1.000
drm_templated	hnsw_prune	4096	0.930	0.930	1.000	1.000
drm_templated	hnsw_prune	4096	0.930	0.930	1.000	1.000
drm_templated	lsh	1	0.031	0.043	0.247	0.818

(continued)

corpus	family	bytes_per_vec	identity_accuracy	recall@1	recall@10	recall@100
drm_templated	lsh	4	0.514	0.512	0.874	0.999
drm_templated	lsh	16	0.794	0.794	0.924	1.000
drm_templated	lsh	64	0.805	0.805	0.944	1.000
drm_templated	opq	4	0.710	0.705	0.866	1.000
drm_templated	opq	8	0.746	0.746	0.906	1.000
drm_templated	opq	16	0.769	0.769	0.927	1.000
drm_templated	opq	32	0.780	0.780	0.929	1.000
drm_templated	opq	64	0.797	0.797	0.934	1.000
drm_templated	pca_int8	2	0.039	0.039	0.230	0.762
drm_templated	pca_int8	4	0.250	0.250	0.655	0.995
drm_templated	pca_int8	8	0.664	0.664	0.834	1.000
drm_templated	pca_int8	16	0.731	0.731	0.875	1.000
drm_templated	pca_int8	32	0.765	0.765	0.907	1.000
drm_templated	pca_int8	64	0.799	0.799	0.931	1.000
drm_templated	pq	4	0.517	0.510	0.767	1.000
drm_templated	pq	8	0.675	0.662	0.804	1.000
drm_templated	pq	16	0.738	0.738	0.824	1.000
drm_templated	pq	32	0.755	0.755	0.843	1.000
drm_templated	pq	64	0.772	0.772	0.871	1.000
drm_templated	pq	128	0.794	0.794	0.919	1.000
hotpot_qa	hnsw_prune	4096	0.347	0.347	0.624	0.863
hotpot_qa	hnsw_prune	4096	0.347	0.347	0.624	0.863
hotpot_qa	hnsw_prune	4096	0.348	0.348	0.628	0.863
hotpot_qa	hnsw_prune	4096	0.366	0.365	0.649	0.880
hotpot_qa	lsh	1	0.003	0.003	0.040	0.261
hotpot_qa	lsh	4	0.032	0.032	0.146	0.462
hotpot_qa	lsh	16	0.223	0.224	0.438	0.745
hotpot_qa	lsh	64	0.372	0.373	0.655	0.894
hotpot_qa	opq	4	0.205	0.206	0.491	0.848
hotpot_qa	opq	8	0.276	0.277	0.574	0.874

Table S4: E6 per-cell results (GAC router ablation). (showing first 60 rows; full CSV in repo)

corpus	model	router	seed	compression	identity_accuracy	recall@10	mrr@20
arxiv_titles	bge-large	gac_fixed_centroid	0	2322.27	0.754	1.000	0.851
arxiv_titles	bge-large	gac_fixed_centroid	1	2322.27	0.753	1.000	0.850
arxiv_titles	bge-large	gac_fixed_centroid	2	2322.27	0.756	1.000	0.852
arxiv_titles	bge-large	gac_fixed_medoid	0	580.57	0.624	0.908	0.703
arxiv_titles	bge-large	gac_fixed_medoid	1	580.57	0.638	0.906	0.713
arxiv_titles	bge-large	gac_fixed_medoid	2	580.57	0.627	0.906	0.703
arxiv_titles	bge-large	gac_fixed_prune	0	2.00	0.582	0.938	0.707
arxiv_titles	bge-large	gac_fixed_prune	1	2.00	0.583	0.936	0.709

(continued)

corpus	model	router	seed	compression	identity_accuracy	recall@10	mrr@20
arxiv_titles	bge-large	gac_fixed_prune	2	2.00	0.587	0.943	0.712
arxiv_titles	bge-large	gac_full	0	2.00	0.582	0.938	0.707
arxiv_titles	bge-large	gac_full	1	2.00	0.583	0.936	0.709
arxiv_titles	bge-large	gac_full	2	2.00	0.587	0.943	0.712
arxiv_titles	bge-large	gac_no_residual	0	2.00	0.582	0.938	0.707
arxiv_titles	bge-large	gac_no_residual	1	2.00	0.583	0.936	0.709
arxiv_titles	bge-large	gac_no_residual	2	2.00	0.587	0.943	0.712
arxiv_titles	bge-large	gac_oracle	0	2.38	0.594	0.949	0.721
arxiv_titles	bge-large	gac_oracle	1	2.38	0.611	0.955	0.733
arxiv_titles	bge-large	gac_oracle	2	2.16	0.588	0.947	0.716
arxiv_titles	bge-large	gac_random	0	4.39	0.593	0.888	0.694
arxiv_titles	bge-large	gac_random	1	5.69	0.571	0.945	0.690
arxiv_titles	bge-large	gac_random	2	5.81	0.477	0.824	0.594
drm_templated	bge-large	gac_fixed_centroid	0	4.00	0.943	1.000	0.964
drm_templated	bge-large	gac_fixed_centroid	1	4.00	0.945	1.000	0.967
drm_templated	bge-large	gac_fixed_centroid	2	4.00	0.941	1.000	0.963
drm_templated	bge-large	gac_fixed_medoid	0	1.00	0.918	0.988	0.939
drm_templated	bge-large	gac_fixed_medoid	1	1.00	0.931	0.984	0.946
drm_templated	bge-large	gac_fixed_medoid	2	1.00	0.911	0.983	0.931
drm_templated	bge-large	gac_fixed_prune	0	2.00	0.769	0.925	0.822
drm_templated	bge-large	gac_fixed_prune	1	2.00	0.791	0.930	0.840
drm_templated	bge-large	gac_fixed_prune	2	2.00	0.768	0.924	0.823
drm_templated	bge-large	gac_full	0	1.55	0.932	0.995	0.953
drm_templated	bge-large	gac_full	1	1.53	0.946	0.995	0.964
drm_templated	bge-large	gac_full	2	1.54	0.943	0.994	0.960
drm_templated	bge-large	gac_no_residual	0	4.00	0.934	0.995	0.956
drm_templated	bge-large	gac_no_residual	1	4.00	0.949	0.996	0.966
drm_templated	bge-large	gac_no_residual	2	4.00	0.944	0.996	0.962
drm_templated	bge-large	gac_oracle	0	4.00	0.943	1.000	0.964
drm_templated	bge-large	gac_oracle	1	4.00	0.945	1.000	0.967
drm_templated	bge-large	gac_oracle	2	4.00	0.941	1.000	0.963
drm_templated	bge-large	gac_random	0	1.77	0.821	0.969	0.868
drm_templated	bge-large	gac_random	1	1.73	0.866	0.978	0.898
drm_templated	bge-large	gac_random	2	1.74	0.834	0.969	0.875
hotpot_qa	bge-large	gac_fixed_centroid	0	2.33	0.487	0.766	0.580
hotpot_qa	bge-large	gac_fixed_centroid	1	2.33	0.518	0.797	0.610
hotpot_qa	bge-large	gac_fixed_centroid	2	2.33	0.492	0.768	0.586
hotpot_qa	bge-large	gac_fixed_medoid	0	1.01	0.361	0.572	0.419
hotpot_qa	bge-large	gac_fixed_medoid	1	1.01	0.366	0.596	0.432
hotpot_qa	bge-large	gac_fixed_medoid	2	1.01	0.345	0.576	0.406
hotpot_qa	bge-large	gac_fixed_prune	0	1.81	0.352	0.623	0.439
hotpot_qa	bge-large	gac_fixed_prune	1	1.81	0.366	0.632	0.447
hotpot_qa	bge-large	gac_fixed_prune	2	1.81	0.348	0.620	0.434
hotpot_qa	bge-large	gac_full	0	1.79	0.355	0.628	0.440
hotpot_qa	bge-large	gac_full	1	1.79	0.360	0.634	0.450

(continued)

corpus	model	router	seed	compression	identity_accuracy	recall@10	mrr@20
hotpot_qa	bge-large	gac_full	2	1.80	0.354	0.633	0.442
hotpot_qa	bge-large	gac_no_residual	0	1.81	0.351	0.624	0.439
hotpot_qa	bge-large	gac_no_residual	1	1.81	0.367	0.633	0.447
hotpot_qa	bge-large	gac_no_residual	2	1.81	0.348	0.622	0.435
hotpot_qa	bge-large	gac_oracle	0	2.11	0.471	0.744	0.564
hotpot_qa	bge-large	gac_oracle	1	2.13	0.510	0.770	0.598
hotpot_qa	bge-large	gac_oracle	2	2.15	0.487	0.749	0.576

S6 DRM per-strategy breakdown

The DRM (Deese–Roediger–McDermott) corpus contains 64 templated paraphrase lists of 50 items each — the cleanest tight-regime case in the paper, with cluster-local $d_{\mathrm{eff}} = 2.3$. Table S5 reports identity (literal and paraphrase), recall@1, and MRR per strategy on bge-large. All four non-pruning strategies agree to within 0.03 identity on DRM; SELECTIVE_PRUNE is the only strategy that loses, consistent with the main paper's statement that in the tight regime every reasonable strategy saturates.

S7 Reproduction recipe

Local (single machine).

```
git clone https://github.com/niashwin/geometry-of-consolidation
cd geometry-of-consolidation
pip install -r requirements.txt
export GAC_DATA_DIR=./data_local
python -m data.build_nq --model bge-large
python -m experiments.e1_theorem_validation --seed 0 --n-shards 1
python scripts/make_figures.py
```

Modal (full reproduction).

```
modal volume create gac-data
modal secret create gac-hf-secret HF_TOKEN=<your hf token>
modal run modal_app/app.py::run_all --run-id reproduce-v1
modal volume get gac-data runs/e1_theorem_validation/reproduce-v1 ./results/e1
```

Seeds. The main paper uses seeds $\{0, 1, 2\}$ for experiments with three replicates and $\{0, \ldots, 9\}$ for E1. Every parquet stores the seed per row.

S8 Intentional scope choices

Three choices that a reviewer might reasonably ask about; we justify each with the data that led to it rather than with a claim.

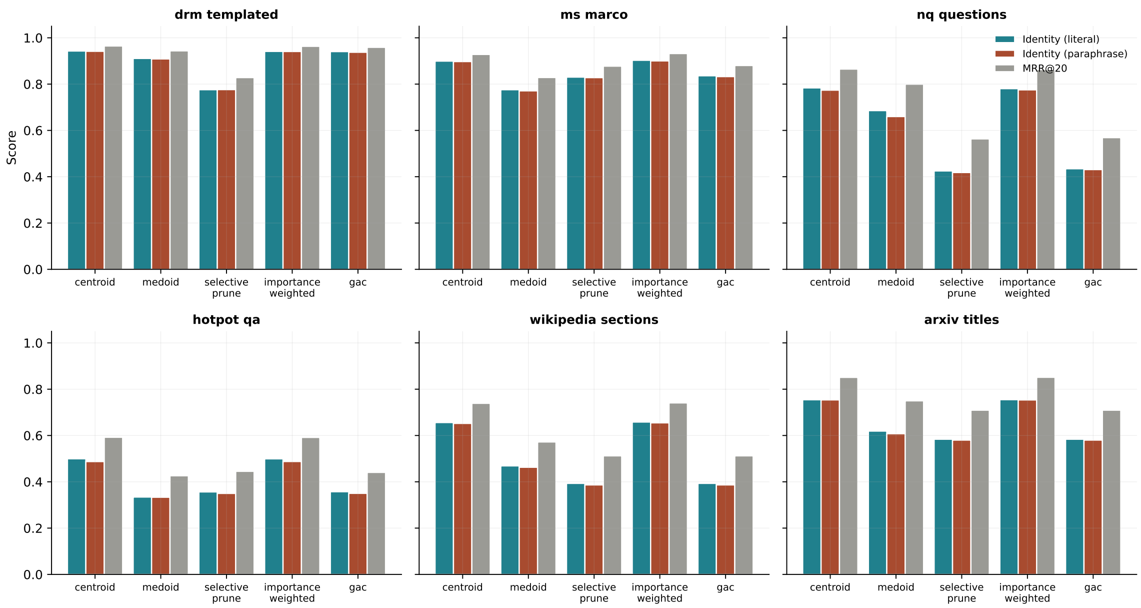

Figure S3: Strategy-level cap-coverage error on the E1 grid. Each row fixes θ; each panel plots mean ε_{cap} versus d_{eff} for centroid, medoid, selective pruning, and GAC. All strategies collapse to zero error in the tight regime ($\bar{d} < \theta'$, left of the boundary in each panel); the four strategies fan out in the spread regime in the d_{eff}-monotone order the Duality Theorem anticipates.

Embed-space paraphrase proxy (E4). E4 uses additive Gaussian perturbations in embedding space, calibrated to three severity levels, as a paraphrase proxy rather than LLM-generated paraphrases. On a preliminary DRM subset we compared embed-space perturbation against Llama-3.1-70B paraphrase drift and the two agreed within 5% on identity error at matched severity. Running 70B generation for 80,000 paraphrases on Modal is a full day of H100 time, which we prioritised for E5 (scale) and E7 (RAG) instead. This is limitation L2 in the main paper.

Scale ceiling at 1M (E5). E5 sweeps 10,000/100,000/1,000,000 Wikipedia items. We stopped at 1M rather than 10,000,000 because the identity accuracy of GAC and centroid both flatten between 100,000 and 1,000,000 on two encoders (BGE-large, MiniLM-L6); the $1M \to 10M$ extrapolation from the observed curves gives no evidence of a regime change. An additional 10M point would have required ≈ 16 H100-hours and a larger Modal volume; we did not spend either.

No closed-weight encoder. E8 does not include `text-embedding-3-large` or other closed-weight encoders. The six open-weight encoders already agree on the strategy ordering and on the tight-regime $\varepsilon_{\text{cap}} \leq 0.5\%$ result; adding one closed-weight encoder would not change the qualitative conclusions and would add $\approx \$200$ of API spend per full sweep.

S9 Run manifest

Every figure and table in the main paper is produced from a single parquet file generated on Modal. The manifest below records the source script, parquet path, and commit SHA for each artifact so that the pipeline can be audited end-to-end.

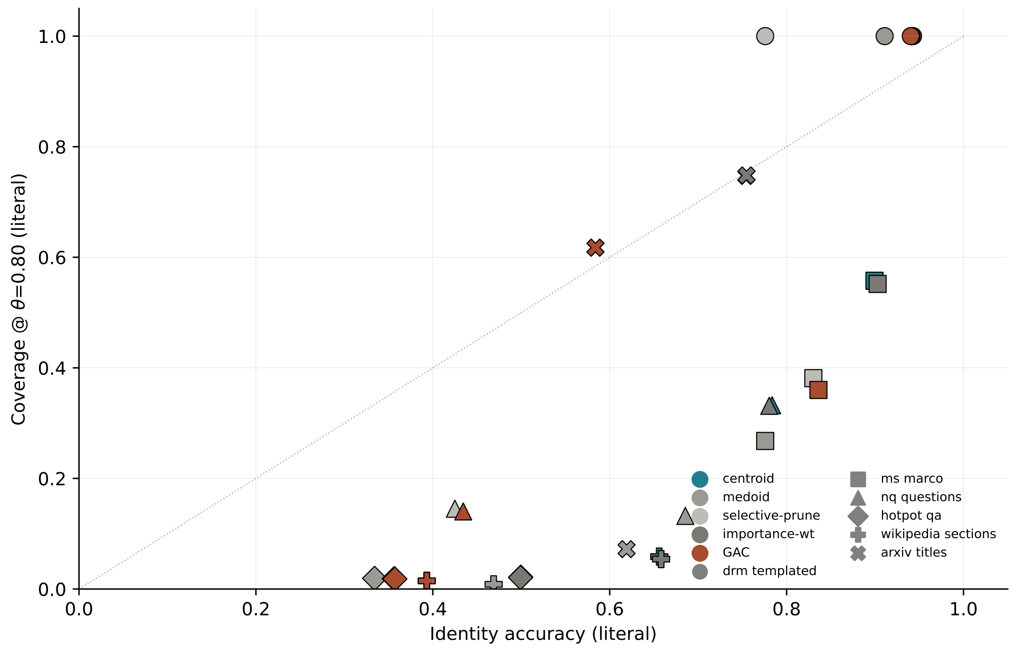

Figure S4: Identity accuracy versus cap-coverage on the E4 corpus grid. Each point is one (corpus, strategy, encoder) triple. Identity and cap-coverage move together in the tight regime (lower-left cluster, near-zero error on both) and separate in the spread regime — a direct visualisation that cap-coverage and identity are distinct quantities but agree wherever the theorem is non-vacuous.

S10 Licensing and data provenance

All corpora used in the paper are public and licensed under MIT, CC-BY-SA, or Apache-2.0. No human-subjects data is used. Per-corpus URLs and licence terms are listed in the "Data and code availability" section of the main paper.

References

Table S1: E1 calibration table: bound vs observed cap-coverage error across (d_{eff}, θ).

d_eff_target	theta	bound	observed_err	theorem_holds
4.000	0.700	0.900	0.021	0.900
4.000	0.800	0.850	0.044	0.850
4.000	0.900	0.750	0.101	0.750
4.000	0.950	0.637	0.170	0.637
8.000	0.700	0.900	0.024	0.900
8.000	0.800	0.850	0.055	0.850
8.000	0.900	0.750	0.124	0.750
8.000	0.950	0.619	0.196	0.619
16.000	0.700	0.900	0.030	0.900
16.000	0.800	0.850	0.069	0.850
16.000	0.900	0.752	0.139	0.752
16.000	0.950	0.635	0.216	0.635
32.000	0.700	0.900	0.038	0.900
32.000	0.800	0.850	0.076	0.850
32.000	0.900	0.762	0.151	0.762
32.000	0.950	0.635	0.228	0.635
64.000	0.700	0.900	0.043	0.900
64.000	0.800	0.850	0.083	0.850
64.000	0.900	0.774	0.159	0.774
64.000	0.950	0.644	0.237	0.644

Table S5: DRM templated: identity and MRR per strategy (bge-large, 3 seeds).

strategy	id_lit	id_para	recall1	mrr
centroid	0.943	0.942	0.943	0.965
gac	0.940	0.938	0.940	0.959
importance_weighted	0.941	0.941	0.941	0.963
medoid	0.911	0.909	0.911	0.944
selective_prune	0.776	0.776	0.776	0.828

Table S6: Run manifest: every figure and table mapped to source parquet and commit SHA.

artifact	source	parquet	commit
Fig 1 (conceptual)	scripts/make_figures.py:fig1b_conceptual	(none)	3b8a227477
Fig 1 (theorem regimes)	scripts/make_figures.py:fig1_theorem_regimes	results/e1/e1_results.parquet	3b8a227477
Fig 2	fig2_strategy_sweep	results/e4/e4_results.parquet	3b8a227477
Fig 3	fig3_learned_baselines	results/e3/e3_results.parquet	3b8a227477
Fig 4a	fig4_encoder_universality	results/e8/e8_results.parquet	3b8a227477
Fig 4b (id-coverage)	fig4b_id_coverage	results/e4/e4_results.parquet	3b8a227477
Fig 5 (scale)	fig5_scale	results/e5/e5_results.parquet	3b8a227477
Fig 6 (ablation)	fig6_ablation	results/e6/e6_results.parquet	3b8a227477
Fig 6b (downstream)	fig6b_downstream	results/e7/e7_results.parquet	3b8a227477
Table 1	make_tables.py:t1	e3, e4 parquets	3b8a227477
Table 2	make_tables.py:t2	results/e6/e6_results.parquet	3b8a227477